

PROBLEMS CLASS 3: SOLUTIONS

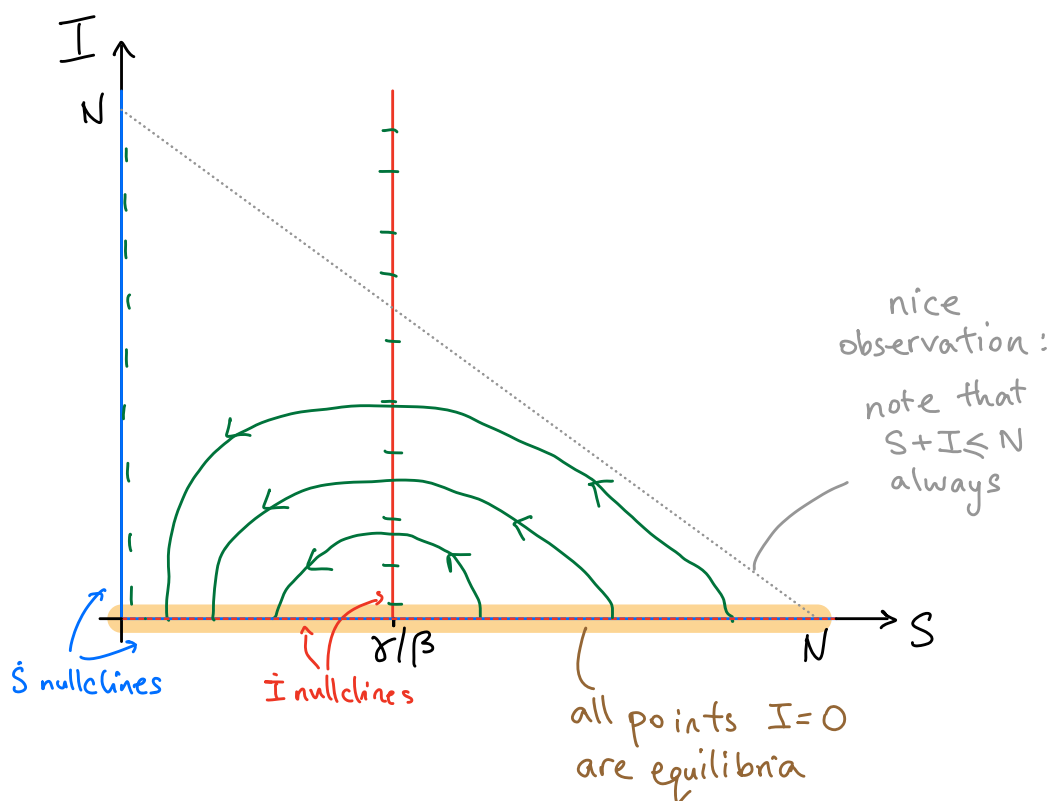
1. SIR model revisited

- (a) S is the susceptible population, I the infected, R the recovered, N the total population. The term βIS represents the number of susceptible–infected interactions, and γI is the recovery rate.

Social distancing should reduce the value of β , as this represents interactions. This term could be time-dependent: if measuring over days, then lockdown surely would decrease β for a certain period of time.

R includes individuals that have both recovered from the disease, and those who have died. In both cases, they can not catch the disease again.

- (b) We had that $R_0 = \beta N / \gamma$.
- (c) The phase plane, if $\gamma / \beta < N$, looks like



Equilibria are all along the line $I = 0$: the first time we've seen this. Equilibria to the left of the I nullcline are stable; to the right, unstable.

- (d) First note that S is bounded above by N (you can't have more susceptible people than people). If the I nullcline is to the right of N , then I is decreasing in the entire permissible space, i.e. the disease is dying out. What does it mean for the nullcline to be to the right of N ? It means

$$\gamma / \beta > N \implies N / R_0 > N \implies R_0 < 1, \quad (1)$$

which is what we already knew from Problems class 1, but this time we see it graphically.

- (e) (i) With no other information, we could assume that the number of patients requiring hospitalisation is a proportion of those who are infected, I .
(ii) We look for when $dI/dt = 0$, i.e.

$$0 = I(\beta S - \gamma). \quad (2)$$

The maximum (assuming that it is a maximum, of course) occurs when $S = \gamma/\beta$.

- (iii) We get

$$\frac{dI}{dS} = \frac{\beta IS - \gamma I}{-\beta IS} = -1 + \frac{\gamma}{\beta S}. \quad (3)$$

Integrating, this gives us

$$I = -S + \frac{\gamma}{\beta} \log S + C, \quad (4)$$

where our initial condition of $I = I_0$, $S = N - I_0$ gives us that

$$C = I_0 + N - I_0 - \frac{\gamma}{\beta} \log(N - I_0), \quad (5)$$

$$= N - \frac{\gamma}{\beta} \log(N - I_0). \quad (6)$$

The maximum occurs when $S = \gamma/\beta$, giving us

$$I_{\max} = -\frac{\gamma}{\beta} + \frac{\gamma}{\beta} \log \frac{\gamma}{\beta} + N - \frac{\gamma}{\beta} \log(N - I_0). \quad (7)$$

Substituting in $R_0 = \beta N/\gamma$,

$$I_{\max} = -\frac{N}{R_0} + \frac{N}{R_0} \log \frac{N}{R_0} + N - \frac{N}{R_0} \log(N - I_0) \quad (8)$$

$$= N \left[1 - \frac{1}{R_0} \left(1 - \log \frac{N/R_0}{N - I_0} \right) \right]. \quad (9)$$

If $I_0 \ll N$, we are left with

$$I_{\max} = N \left[1 - \frac{1 + \log R_0}{R_0} \right]. \quad (10)$$

2. SIR with temporary immunity

- (a) Our new system is

$$\frac{dS}{dt} = -\beta IS + \delta R, \quad (11)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (12)$$

$$\frac{dR}{dt} = \gamma I - \delta R, \quad (13)$$

$$N = S + I + R. \quad (14)$$

(b) In looking for equilibria, we set $dS/dt = dI/dt = dR/dt = 0$:

$$0 = -\beta IS + \delta R, \quad (15)$$

$$0 = I(\beta S - \gamma), \quad (16)$$

$$0 = \gamma I - \delta R, \quad (17)$$

$$N = S + I + R. \quad (18)$$

Equation (16) tells us that one possibility is $I = 0$, in which case it follows that $R = 0$ and $S = N$ from the other equations: i.e. this is our disease-free equilibrium.

It also gives us the other possibility that $S = \gamma/\beta$. In which case, replacing R with $N - S - I = N - \gamma/\beta - I$ in eq. (15) gives us

$$0 = -\beta I \frac{\gamma}{\beta} + \delta \left(N - \frac{\gamma}{\beta} - I \right) \quad (19)$$

$$= -I\gamma + \delta \left(N - \frac{\gamma}{\beta} - I \right) \quad (20)$$

$$= -I(\gamma + \delta) + \delta \left(N - \frac{\gamma}{\beta} \right), \quad (21)$$

i.e.

$$I = \frac{\delta(N - \gamma/\beta)}{\gamma + \delta} = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\gamma}{\delta}}. \quad (22)$$

Then,

$$R = N - S - I \quad (23)$$

$$= N - \frac{\gamma}{\beta} - \frac{\delta(N - \frac{\gamma}{\beta})}{\gamma + \delta} \quad (24)$$

$$= \frac{N(\gamma + \delta) - \frac{\gamma}{\beta}(\gamma + \delta) - \delta(N - \frac{\gamma}{\beta})}{\gamma + \delta} \quad (25)$$

$$= \frac{N\gamma - \gamma^2/\beta}{\gamma + \delta} = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\delta}{\gamma}}. \quad (26)$$

This is different to our original SIR model because although the disease-free equilibrium is the same, we don't have an additional one-parameter family of equilibria.

(c) As instructed,

$$\frac{dS}{dt} = -\beta IS + \delta(N - S - I), \quad (27)$$

$$\frac{dI}{dt} = \beta IS - \gamma I. \quad (28)$$

The Jacobian, $\mathbf{J}$, is

$$\mathbf{J} = \begin{pmatrix} -(\beta I + \delta) & -(\beta S + \delta) \\ \beta I & \beta S - \gamma \end{pmatrix}. \quad (29)$$

- (d) Linear stability analysis: on our disease-free equilibrium, $(S, I, R) = (N, 0, 0)$, our Jacobian is

$$\mathbf{J} = \begin{pmatrix} -\delta & -(\beta N + \delta) \\ 0 & \beta N - \gamma \end{pmatrix}. \quad (30)$$

Since there is a zero on the off-diagonal, the eigenvalues are just the terms on the diagonal, namely

$$\lambda_1 = -\delta, \quad \lambda_2 = \beta N - \gamma. \quad (31)$$

The second eigenvalue is only positive when $\beta N / \gamma > 1$, i.e. $R_0 > 1$, in which case this is *unstable*. If $R_0 < 1$, it is *asymptotically stable*.

[If you did this the trace/determinant way—and you shouldn't, because the eigenvalues can be read off directly—you would get

$$\text{tr}(\mathbf{J}) = \beta N - \gamma - \delta, \quad \det(\mathbf{J}) = -\delta(\beta N - \gamma). \quad (32)$$

The requirement for stability of positive determinant and negative trace means

$$\det(\mathbf{J}) > 0 \implies \beta N - \gamma < 0, \quad (33)$$

$$\text{tr}(\mathbf{J}) < 0 \implies \beta N - \gamma - \delta < 0, \quad (34)$$

but since the bottom condition is weaker than the top, we just have $\beta N - \gamma < 0$, i.e. $R_0 < 1$ for asymptotic stability.]

As for the other equilibrium where $S = \gamma / \beta$, no off-diagonal zeros so we go for the trace/determinant approach directly:

$$\text{tr}(\mathbf{J}) = -(\beta I + \delta) + \beta S - \gamma \quad (35)$$

$$= -(\beta I + \delta) \quad (36)$$

which is negative because β , I and δ are all positive terms. As for the determinant,

$$\det(\mathbf{J}) = -(\beta I + \delta)(\beta S - \gamma) + (\beta S + \delta)\beta I \quad (37)$$

$$= (\gamma + \delta)\beta I \quad (38)$$

which is positive because it consists of purely positive terms. Since stability requires positive determinant and negative trace, this equilibrium is *asymptotically stable*.

- (e) (I leave this up to you)

3. Advection and diffusion

These questions are here just to solidify your intuition.

- (a) (for example) Diffusion is the spreading out of the population over space, essentially due to individual random motion (but the centre of mass stays the same).
- (b) (for example) Advection is the movement of the centre of mass of the population, or in other words a bulk motion (think ducks being carried away down a river).
- (c) $f(c)$ is any source or sink term, and this can be any of the population models we have seen so far this term.
- (d) In one spatial dimension, this is

$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial x}(cv) + D\frac{\partial^2 c}{\partial x^2} + f. \quad (39)$$